Nasrul Huda

MSc Data Science and Artificial Intelligence

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Professional Summary

Aspiring Data Scientist and AI Engineer with a strong foundation in statistics, machine learning, and modern data technologies. Maths enthusiast committed to translating complex problems into clear, actionable insights. Passionate about building robust, scalable systems and continuously exploring new tools and methods. Also engaged in freelancing, delivering practical, well-designed solutions tailored to client needs.

Education

University of Hamburg Hamburg, Germany
*Master of Science in Data Science and Artificial Intelligence - **Grade - 1,8*** *Oct 2024 - Present*

Jamia Millia Islamia New Delhi, India
*Bachelor of Technology in Computer Science Engineering **Grade - 1,6*** *Oct 2020 - Jun 2024*

Experience

SynTwin GmbH Munich, Germany
AI Engineer *Oct 2025 - Present, Part Time*

- Developing and maintaining the SynTwin platform, a platform for creating and interacting with Digital Twins.
- Currently responsible for developing Agentic-AI workflows.
- **Technical Skills:** Python, Livekit, Supabase, Docker, Git

University of Hamburg - Security and Distributed Systems Research Group Hamburg, Germany
Student Research Assistant *July 2025 - Sept 2025, Part Time*

- Developing Python-based tools and applications for research data management and security analysis.
- Contributing to the NFDIxCS (National Research Data Infrastructure for Computer Science) project, collaborating on interdisciplinary projects involving AI and data science methodologies.
- **Technical Skills:** Python Development, Research Methodology, Cybersecurity, Network Security, Distributed Systems, Docker, Git, Linux

Pettoo UG Hamburg, Germany
Freelance Software Engineer *June 2025 - September 2025, Part Time*

- Developing and maintaining the Pettoo platform, a platform for pet owners to find and book pet services, buy pet products, and connect with other pet owners.
- Currently responsible for the development of the backend of the platform and managing a team of 3 developers, the database structure, and implementing the business logic.
- **Technical Skills:** FastAPI, Google Cloud Platform, PostgreSQL, Docker, Git, Unix, Github Actions, CI/CD, Celery, Elasticsearch

University of Hamburg - Language Technology Research Group Hamburg, Germany
Project Student Assistant *Feb 2025 - May 2025, Part Time*

- Contributed to the development of the DATS Project (Discourse Analysis Tool Suite), an advanced NLP platform for automated text analysis and discourse processing, implementing key features and functionality enhancements.
- Developed and maintained full-stack components including React TypeScript frontend interfaces and FastAPI backend services, implementing RESTful APIs and database integrations for improved system performance and user experience.
- Implemented NLP processing modules, text preprocessing pipelines, and data analysis algorithms, working with machine learning frameworks to handle large-scale linguistic datasets and generate meaningful insights.
- **Technical Skills:** Natural Language Processing, Machine Learning, React TypeScript, FastAPI, Python, RESTful APIs, Database Integration, Docker, Git, Agile Development, Text Analytics, Data Processing

Projects

Few-Shot Bioacoustic Classification | Python, PyTorch Lightning, Prototypical Networks, Hydra, MLflow

- Implemented prototypical networks for animal vocalization classification that adapts to new species from just a few labeled examples using episodic N-way, K-shot training paradigm
- Built end-to-end audio processing pipeline with offline feature extraction (STFT → Mel filterbank → Log/PCEN spectrograms) and online episodic training using PyTorch Lightning with configurable Hydra parameters and MLflow experiment tracking

Federated Fine-Tuning of Nicheformer | Python, PyTorch, Federated Learning

- Implemented federated learning framework for fine-tuning Nicheformer transformer model on mouse brain spatial transcriptomics data, comparing centralized (94.45%), federated (92.76%), and local training strategies
- Built data pipeline for anatomical non-IID partitioning (dorsal/mid/ventral regions) with 24-class cell type classification on 250-dimensional features (248 genes + spatial coordinates) achieving competitive 92.76% accuracy in federated setting

GitGood | Python, RAG, GitHub API, OpenAI API

- Built a comprehensive web application that integrates with GitHub via OAuth to analyze and visualize repository data, commit history, and code differences.
- Integrated AI-powered chat assistant using custom RAG system to answer contextual questions about code changes and repository evolution.

ACL Anthology RAG | Python, RAG, FastAPI, React, Qdrant, Langchain

- Built an intelligent research assistant that retrieves papers from 40,000+ ACL papers using dense vector embeddings and LLM-based query reformulation, enabling semantic understanding beyond keyword matching
- Designed and implemented end-to-end system with FastAPI backend (Python), React frontend, Qdrant vector database, and LangChain LLM integration with real-time streaming UI and inline citations

MyTorch | Python, NumPy, Deep Learning, Neural Networks

- Developed a minimal deep learning library inspired by PyTorch, implementing fundamental neural network operations using only NumPy as the Python library dependency, building everything from scratch with fully connected linear layers for educational and prototyping purposes.
- Built core functionalities including forward and backward propagation, gradient computation, and optimization algorithms to facilitate understanding of deep learning fundamentals.

Momentum | Web Scraping, Content Analysis, Research Tools, Machine Learning, Python

- Developing a research assistant web application inspired by Anara Labs' Anara, designed to streamline academic and technical research workflows.
- Building automated web content extraction and organization systems to gather, process, and categorize research materials from multiple online sources.
- Implementing intelligent summarization algorithms to condense and present research findings in structured, actionable formats for enhanced productivity.

Technical Skills

Languages : Python, C++, TypeScript, Javascript, SQL

Relevant courses : Machine Learning, Data Analytics, Natural Language Processing, Data Structures & Algorithms, Databases and Information Systems, Statistical Signal Processing, Language Technology, Computer Vision, Neural Networks

Technologies : PyTorch, PyTorch Lightning, Hydra, Scikit-Learn, Pandas, PostgreSQL, Langchain, Docker, Tableau, Seaborn, FastAPI, Supabase, Redis, React, NodeJs, Github Actions, Google Cloud, BeautifulSoup4, Git, Tailwind Css, MS Office

Languages

English: C1 (Advanced) | **Hindi:** Native | **Urdu:** Native | **German:** A2 (Elementary)

Awards & Achievements

Letter of Excellence and Acknowledgment, Received from the Vice Chancellor of Jamia Millia Islamia.

IELTS 8.0 Band CEFR C1. 9-Reading, 8-Listening, 7.5-Writing, 7.0-Speaking

Achieved 1st Rank in the university for the 4th year of Bachelors (2024).